GENERATIVE FLOW-MATCHING MODELING OF THE NEURODEVELOPMENTAL CONNECTOME VIA DYNAMIC HYPERGRAPHS

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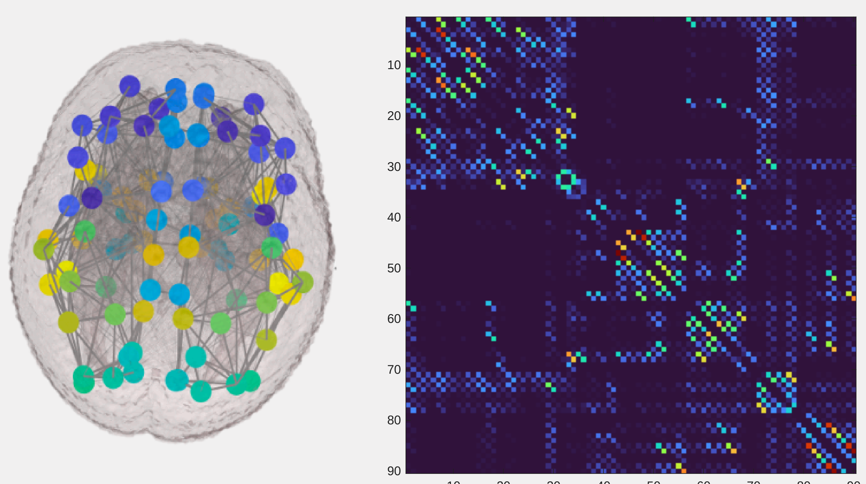
INTRODUCTION

The human brain undergoes an intensive period of non-linear development during the final stages of pregnancy, a trajectory that is disrupted in infants born prematurely.

CHALLENGES

- Structural infant brain imaging datasets are small and expensive to acquire.
- Longitudinal developmental data are particularly sparse across gestational ages.
- Existing generative models often fail to preserve biologically plausible network topology and often regress toward average connectivity patterns.
- Most methods cannot reliably interpolate across unseen developmental stages.



Coordinates and adjacency matrix from "Developing Human Connectome Project" (dHCP) dataset

KEY CONTRIBUTIONS

- Flow-matching framework conditioned on continuous gestational age.
- Dynamic hypergraph and multi-hop transformer layers that learn higher-order dependencies in the brain

About me

Paper

METHODOLOGY

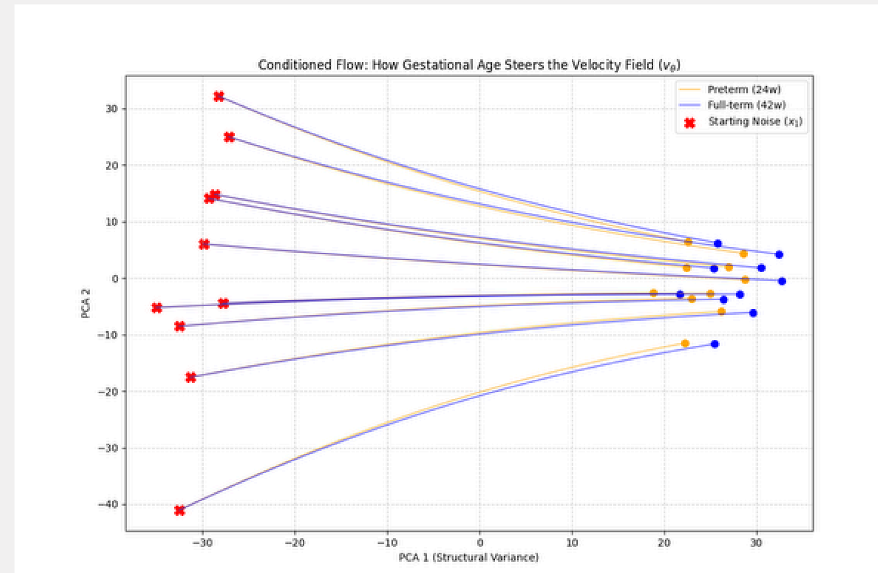
The model conceptualises neurodevelopment as a continuous trajectory rather than discrete snapshots or averages

- Conditional Flow Matching (CFM) learns a vector field to draw a linear probability path between Gaussian noise and the brain connectivity matrix at each age
- 90 brain regions are defined as nodes, in coordinate space, connections between regions are defined as edges, with varying strength.

CFM:
Learns the age-conditioned velocity field.

$$\frac{dx}{dt} = v_\theta(x_t, t, c)$$

FLOW TRAJECTORY

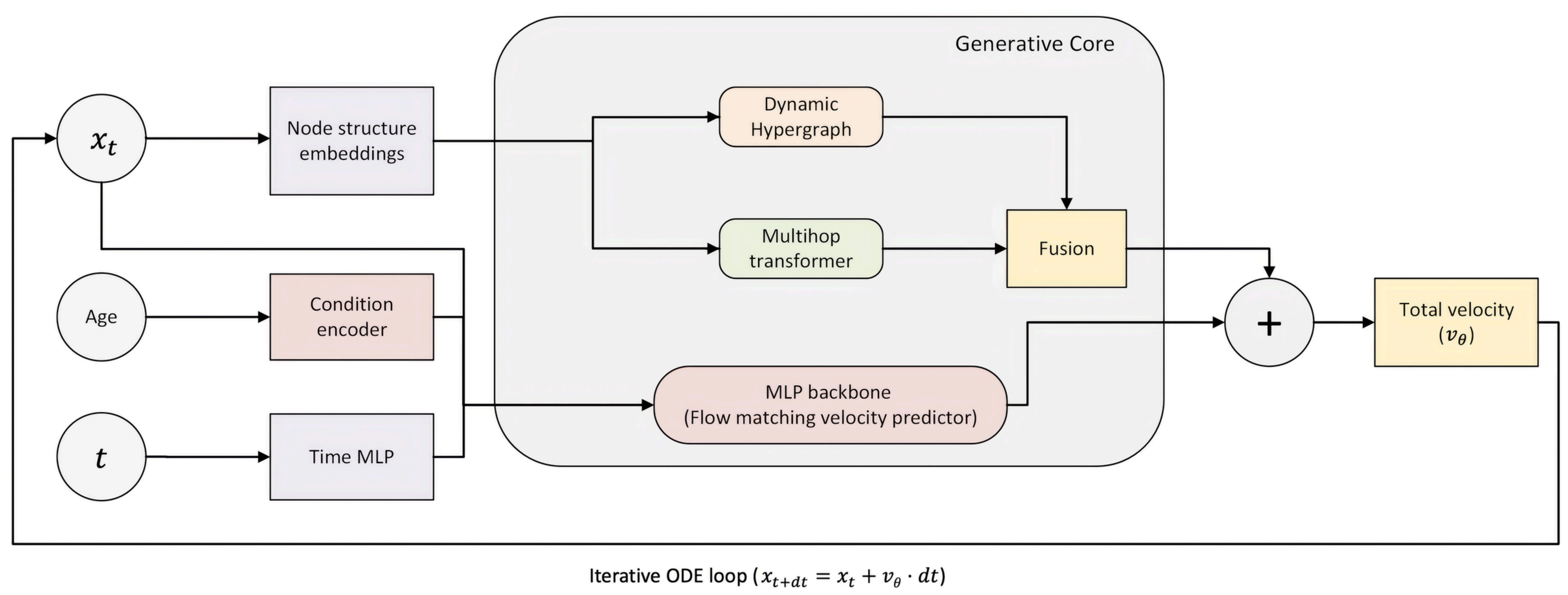


ARCHITECTURAL COMPONENTS VS. VALIDATION LOSS.

Model Name	Architecture		Val. loss
	Transformer	Hypergraph	
Full	✓	✓	**0.148**
Multihop	✓	×	0.432
Hypergraph	×	✓	0.194
Baseline MLP	×	×	0.434

Dynamic Hypergraph
- Uses soft-assignments, allowing nodes and edges to belong to multiple latent clusters that evolve with graph state.

Multihop transformer
- Allowing brain regions to attend to their M-hop topological neighbours.



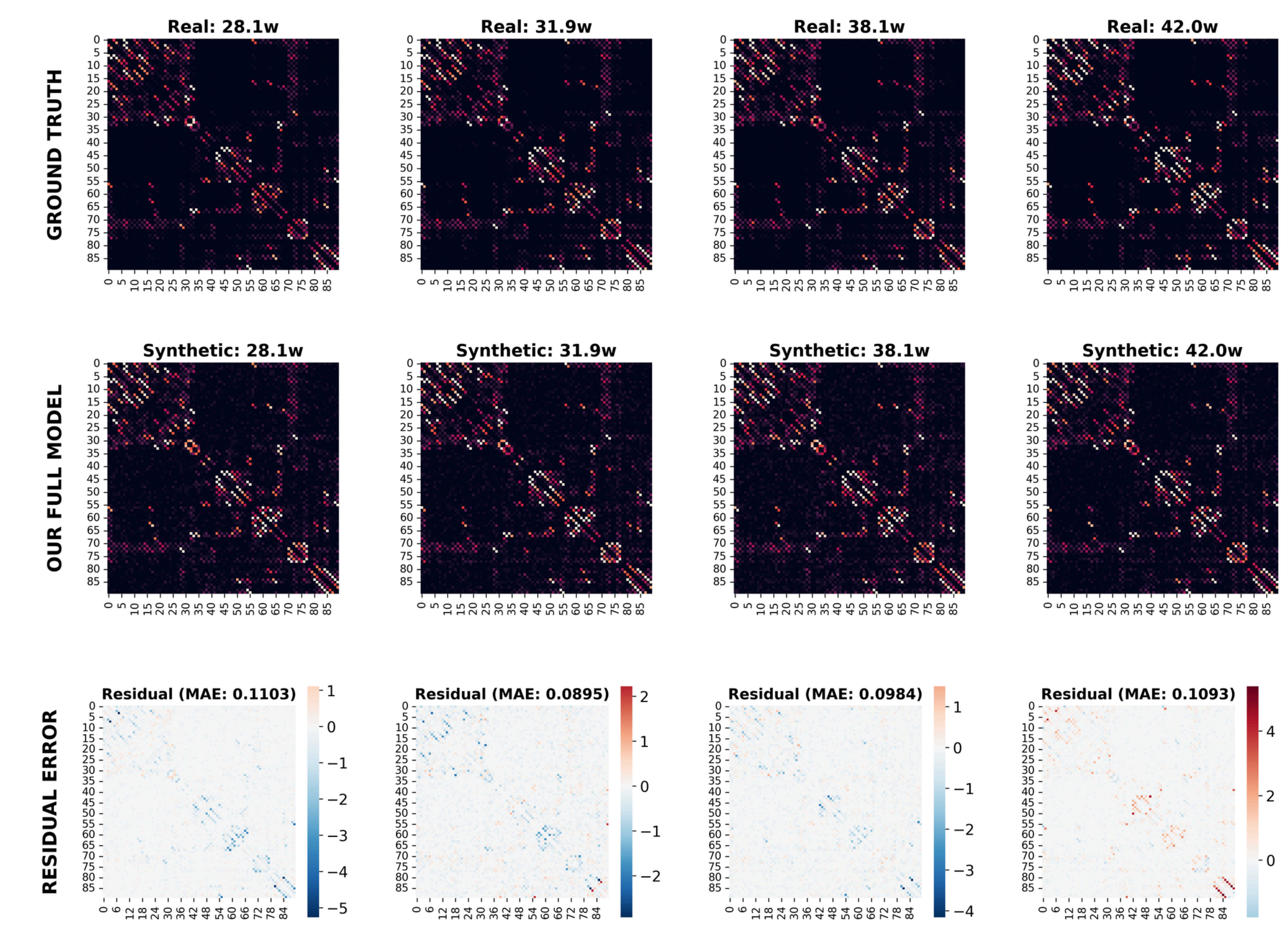
Fusion
- Provides the MLP backbone with both local and global topological context to predict the velocity field
- Updates via ODE solver

CONCLUSIONS

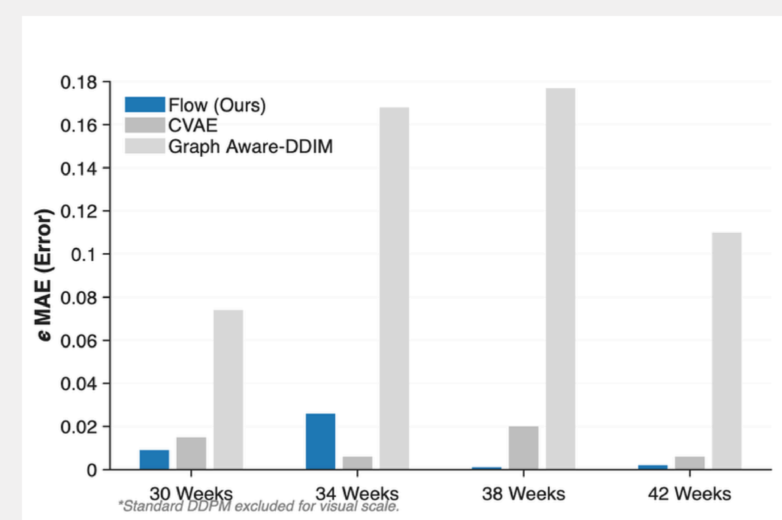
- **Continuous Trajectory Modeling:** Learns brain connectivity as a continuous developmental trajectory, enabling interpolation across unseen gestational ages.
- **Capturing Self-Organization:** Captures evolving higher-order interactions through dynamic hypergraphs and multi-hop transformers.
- **Future Clinical Framework:** Improves downstream prediction by augmenting scarce neurodevelopmental datasets with realistic synthetic connectomes.

RESULTS

GENERATIVE FIDELITY ACROSS THE GESTATIONAL AGE SPECTRUM

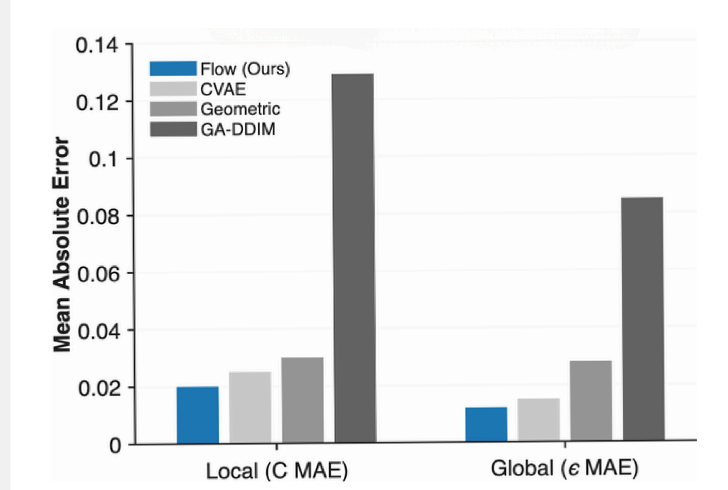


- Heatmaps compare real and synthetic structural connectomes at extreme preterm (28.1w) through to full term (42.0w).
- The residual error heatmaps show near-white areas, demonstrating near-perfect reconstruction with only slight over/underestimations of connection strength.



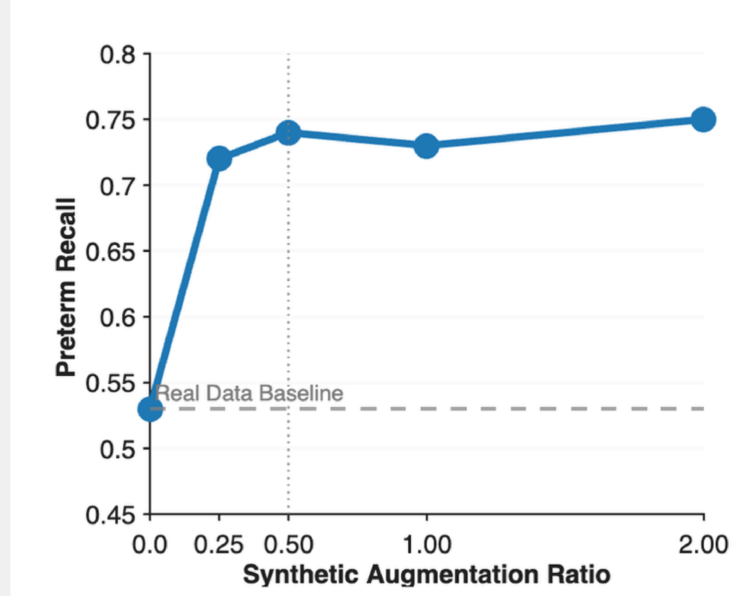
ZERO-SHOT AGE INTERPOLATION

- Outperforms diffusion baselines across all held-out ages (30, 34, 38, and 42 weeks).
- Low MAE confirms the model learns continuous developmental trajectories instead of memorizing training data.



TOPOLOGICAL FIDELITY

- Accurately reproduces complex network features characteristic of real brain data.
- Achieves lowest error for clustering coefficient (C) and global efficiency, preserving critical small-world connectivity.



CLINICAL DOWNSTREAM TASK

- Unlike standard CVAE models that suffer total mode-collapse (Preterm F1: 0.00), our model preserves minority class representation.
- Augmenting real data with synthetic data (optimal 0.5 ratio) maximizes preterm F1 and ROC-AUC, proving real-world clinical utility.

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